# State Machines – Part 1
K. Sridharan

# Review

- We have studied flip-flops, counters, registers etc.
- We will now look at various aspects relating to (i) analysis of sequential circuits and (ii) design of sequential circuits.
- We will begin with related terminology.

# The notion of state

- Recall the use of this term earlier. For a sequential circuit, the contents of the storage elements represent the state of the circuit.
- The behaviour of a sequential circuit is determined from the inputs, outputs and the states of the flip-flops.

# What does analysis entail ?

- Given a logic circuit (typically a picture**), the analysis of sequential circuits involves obtaining tables and graphs for describing the time sequence of inputs, outputs and flip-flop states.

** One can also draw the circuit (given some equations) and then perform analysis.

# What does design involve ?

- Given the desired relationship between the input and output sequences, design involves coming up with the logic circuit in general.

# Tools

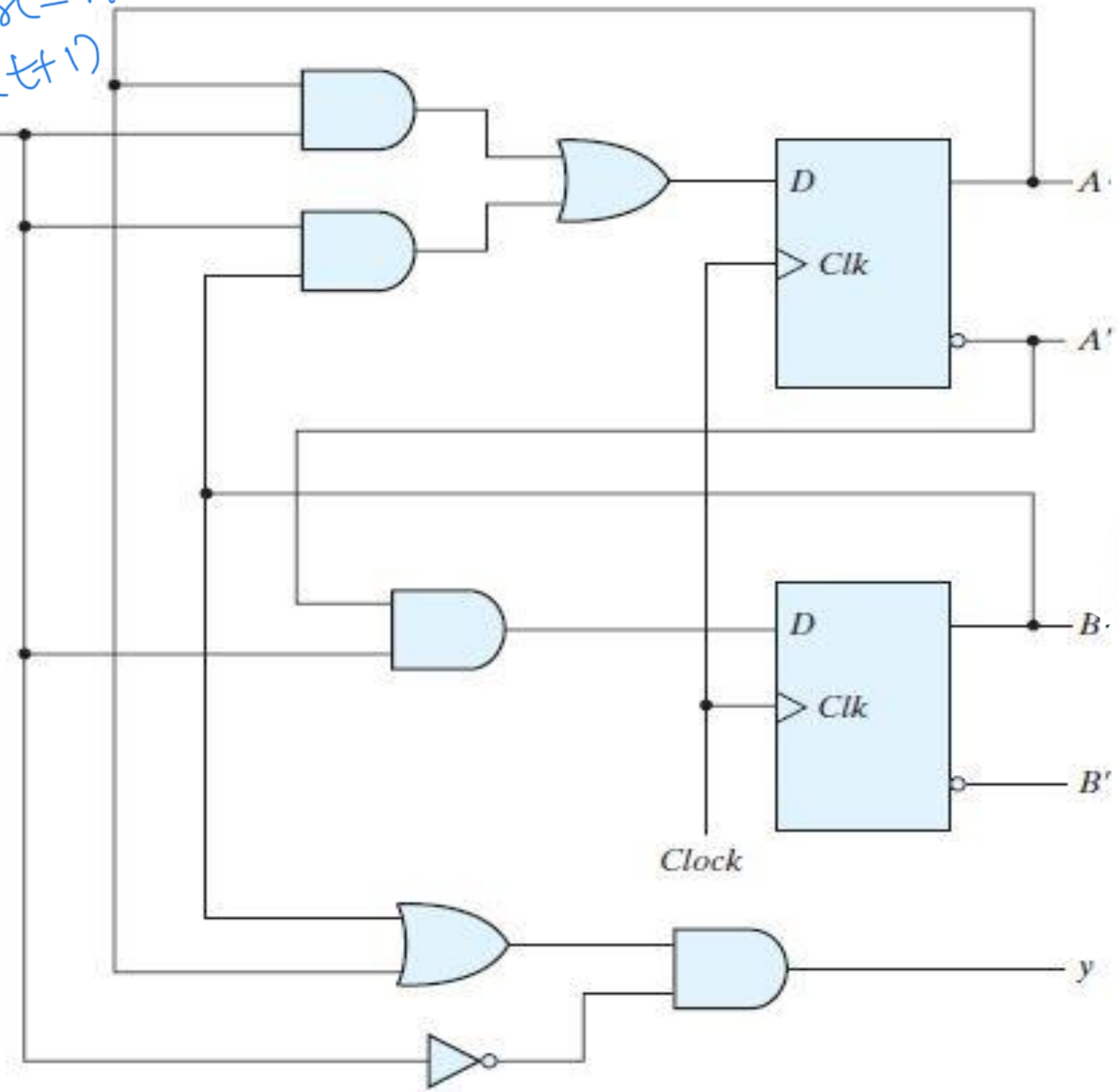
- A number of tools become important in the process of analysis and design.
- These include state equations, state tables, state diagrams, Mealy machine, Moore machine etc.
- We will introduce some of these in this set of slides. We will start with a simple circuit and discuss the various notions.

# Sample Sequential Circuit

$D_A = Ax + Bx = A(t+1)$

$D_B = \bar{A}x = B(t+1)$

$y = (A+B)\bar{x}$



## State equations (or Transition equations)

- The equations governing this circuit are:

$$A(t + 1) = A(t)\,x(t) + B(t)\,x(t)$$

$$B(t + 1) = A'(t)\,x(t)$$

The left-hand side denotes the next state of a flip-flop after a clock edge while the right-hand side specifies the present state and input conditions that make the next state equal to 1.

# Equation for the output y

- $y(t) = [A(t) + B(t)]\, x'(t)$

# Explanation

- The equations have been written directly from the circuit.
- For the state equations, look at the flip-flop outputs.
- For the output y, see the gates at the bottom and their inputs.

# Can also give a state table

- The time sequence of inputs, outputs and flip-flop states can be expressed via a state table (also called as a state transition table).
- The state table for the circuit shown will look like the one given next.

# State table

$y(t) = [A(t) + B(t)]\,\bar{x}$

$A(t+1) = [A(t) + B(t)] \cdot x$ $\quad$ $B(t+1) = A\bar{(t)} \cdot x$

| Present State | | Input | Next State | | Output |
|---|---|---|---|---|---|
| $A(t)$ | $B(t)$ | $x$ | $A(t+1)$ | $B(t+1)$ | $y$ |
| 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 | 1 | 0 |
| 0 | 1 | 0 | 0 | 0 | 1 |
| 0 | 1 | 1 | 1 | 1 | 0 |
| 1 | 0 | 0 | 0 | 0 | 1 |
| 1 | 0 | 1 | 1 | 0 | 0 |
| 1 | 1 | 0 | 0 | 0 | 1 |
| 1 | 1 | 1 | 1 | 0 | 0 |

# Alternate form of State table

| Present State | | Next State | | | | Output | |
|---|---|---|---|---|---|---|---|
| | | $x = 0$ | | $x = 1$ | | $x = 0$ | $x = 1$ |
| $A$ | $B$ | $A$ | $B$ | $A$ | $B$ | $y$ | $y$ |
| 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |
| 0 | 1 | 0 | 0 | 1 | 1 | 1 | 0 |
| 1 | 0 | 0 | 0 | 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 0 | 1 | 0 | 1 | 0 |

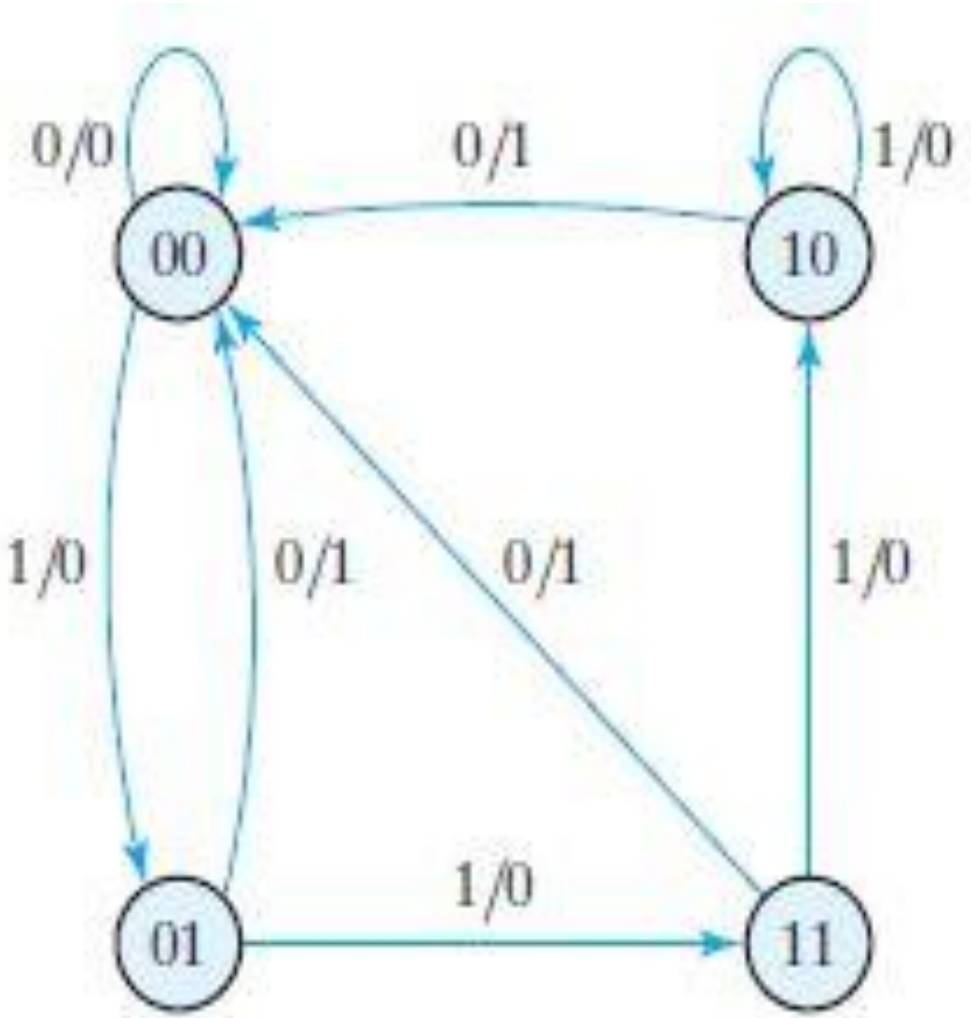
# A note on inputs

- In clocked sequential circuits (such as the one shown in the slides earlier), state transitions are initiated by a clock edge.
- The flip-flops remain in their present states if no clock is applied.
- As a convention, the clock does not appear explicitly as an input variable in the state table.

# Another tool

- Besides state tables, we will also use something called as a state diagram to describe a sequential circuit.
- The state diagram for the circuit shown earlier is given next.

# State Diagram



State in Circle

Input /Output → connecting

# Explanation

- A state is represented by a circle in the state diagram. The binary numbers inside each circle indicate the state of the flip-flops.
- Clock-triggered transitions between the states (such as 00, 01 etc.) are indicated by directed lines connecting the circles. The labels along these lines are to be interpreted as follows: the input value during the present state is labelled first while the number after the slash gives the output during the present state with the given input.

# Example

- Along the directed line from state 00 to state 01, we have a label of 1/0 : this indicates that when the sequential circuit is in the present state (00) and the input (x) is 1, the output (y) is 0. After the next clock cycle, the circuit goes to the next state, namely 01.
- If the input (x) changes to 0, the output (y) becomes 1. Note that the bit value mentioned for the output along a directed line occurs during the present state and with the indicated input and this does not have anything to do with the transition to the next state.

# Clock not shown in state diagram

- As discussed earlier, clocked sequential circuits will have state transitions initiated by a clock edge.
- However, the clock does not explicitly appear as an input variable in the state diagram.

# Textbook and Reference

- M. Mano and M. Ciletti, Digital Design, Pearson Education, 2018
- S. Brown and Z. Vranesic, Fundamentals of digital logic with VHDL design, McGraw Hill, 2017.